RACHANA SANTHEBACHAHALLI UMESHA

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Personal Profile

Electronics and Computer Engineering graduate (MEng, Dublin City University; BE Electronics and Communication, First Class with Distinction) with hands-on experience in circuits, sensors, microcontrollers, substation monitoring systems and energy system decarbonisation. Experienced Lab Demonstrator in Digital and Analog Systems at DCU and a permanent employee of Dublin Bus (CIÉ Group), seeking a graduate role in electrical, electronics or control engineering.

Education & Qualifications

2025 – 2026 **MEng in Electronics and Computer Engineering, Dublin City University**
Expected grade 2.1

Modules include: Energy System Decarbonisation, Security for IoT and Edge Computing, Network Performance, Real-Time Digital Signal Processing, Mathematics, Digital and Analog Systems

Thesis: Packet Spraying Optimisation for Multi-Tier Spine-Leaf and Fat-Tree Networks (supervisor: Dr Martin Collier)

2020 – 2024 **BE in Electronics and Communication Engineering, Maharaja Institute of Technology, India**
First Class with Distinction (CGPA 8.18/10, 74.30%)

Modules include: Basic Electrical Engineering, Network Analysis, Analog Electronic Circuits, Digital System Design, Control Systems, Signals and Systems, Power Electronics, Microcontrollers, Linear Integrated Circuits, Electronic Instrumentation, Digital Signal Processing, VLSI Design, Embedded Systems

Technical Skills

Technical and Programming Skills:
SCADA, PLC Logic (Interlocks, Protection), Modbus TCP, HMI Design, IDMT Overcurrent Protection, Substation Monitoring, Power Systems, Power Electronics, PLCC, Control Systems (PID), Instrumentation (CT/PT), Analog and Digital Circuit Design, MQTT, IoT Security, MATLAB, Python, C/C++, VHDL, SystemVerilog, NS-3, Linux, Microsoft Office

Hardware Skills:
Arduino Uno, ESP8266, Microcontrollers, Sensor Interfacing, Oscilloscope, Multimeter, Function Generator, DC Power Supply, Breadboarding

Relevant Work Experience

Sept 2025 – May 2026: Lab Demonstrator, Digital and Analog Systems, Dublin City University

- Supported students in building and testing analog and digital circuits using lab instruments
- Ran student vivas, graded lab sessions and helped with exam viva grading

Projects

Substation SCADA System (Personal Project, 2026) – github.com/RachanaSU/substation-scada

- Built a SCADA system for an 11 kV / 415 V substation in Python: PLC with a 50 ms scan cycle, Modbus TCP implemented from scratch, and a web HMI with live single-line diagram, breaker control, alarm management and historian

- Programmed earth-switch and reclose interlocks, IEC Standard Inverse IDMT overcurrent and transformer thermal protection; IDMT tripped in 1356 ms against 1363 ms theory, within one PLC scan

Substation Monitoring using IoT (Personal Project, 2026) – github.com/RachanaSU/substation-iot-monitoring

- Simulated IoT monitoring of an 11 kV / 415 V, 500 kVA substation in Python: an edge node publishes current, voltage, transformer oil temperature and DC battery data as JSON over MQTT
- Built a cloud monitor with limit alarms, a rate-of-rise early warning and hysteresis; in a 24-hour fault test it caught an overload, a transformer over-temperature (warned 3.5 hours early) and a battery charger failure

Substation Monitoring using Power Line Carrier Communication (Personal Project, 2026) – github.com/RachanaSU/plcc-substation-simulation

- Simulated a PLCC link in Python sending substation readings over a power line using FSK on a 90/110 kHz carrier, with 50 Hz mains interference and line noise
- Recovered the data with band-pass filtering and demodulation, measured bit error rate against SNR, and raised fault alarms at the control room

Packet Spraying Optimisation (MEng Thesis, DCU) – github.com/RachanaSU/EEN1095_Rachana-S-U

- Compared packet spraying with PT-ECMP, flowlet switching and LetFlow on spine-leaf and fat-tree networks in NS-3 (C++20, Linux); showed packet spraying reorders up to 74% of packets while the alternatives cut reordering to 0% and jitter by over 20x

GPS-Based Accident Detection System (BE)

- Developed an Arduino-based embedded system that detects a vehicle accident from sensor input and reports its GPS location; handled circuit design, sensor interfacing and hardware testing

Other Work Experience

2025 – Present (p/t): Sales Promoter, DoDublin, Dublin Bus (CIÉ Group)

- Permanent employee; named Top Performer 2025
- Maintain Excel trackers in the commercial department and handle customer emails and queries

July – Dec 2024: R&D Intern, UNIABROAD

- Delivered R&D initiatives with cross-functional teams

Jan – June 2022: Web Development Intern, Compsoft Technologies

- Built the front end for the Leaf Now project in HTML, CSS and JavaScript

Interests and Achievements

- Top Performer 2025, DoDublin sales team
- Media Manager, IEEE MITM Student Society (2022 – 2024): ran the society's social media and publicised events
- TCS National Qualifier Test (numerical, verbal and reasoning ability): 78% overall
- Fluent in English and Hindi

Referees

Available on Request